15 Power Supplies (Voltage Regulators)

CHAPTER OBJECTIVES

- How power supply circuits operate
- Operation of RC filters
- Discrete voltage regulator operation
- About practical IC voltage regulators

15.1 INTRODUCTION

Chapter 15 introduces the operation of power supply circuits built using filters, rectifiers, and then voltage regulators. (Refer to Chapter 2 for the initial description of diode rectifier circuits.) Starting with an ac voltage, we obtain a steady dc voltage by rectifying the ac voltage, then filtering to a dc level, and, finally, regulating to obtain a desired fixed dc voltage. The regulation is usually obtained from an IC voltage regulator unit, which takes a dc voltage and provides a somewhat lower dc voltage, which remains the same even if the input dc voltage varies or the output load connected to the dc voltage changes.

A block diagram containing the parts of a typical power supply and the voltage at various points in the unit is shown in Fig. 15.1. The ac voltage, typically 120 V rms, is connected to a transformer, which steps that ac voltage down to the level for the desired dc output. A diode rectifier then provides a full-wave rectified voltage, which is initially filtered by a basic capacitor filter to produce a dc voltage. This resulting dc voltage usually has some ripple or ac voltage variation. A regulator circuit can use this dc input to provide a dc voltage that not only has much less ripple voltage, but also remains at the same dc value even if the input dc voltage varies somewhat or the load connected to the output dc voltage changes. This voltage regulation is usually obtained using one of a number of popular voltage regulator IC units.

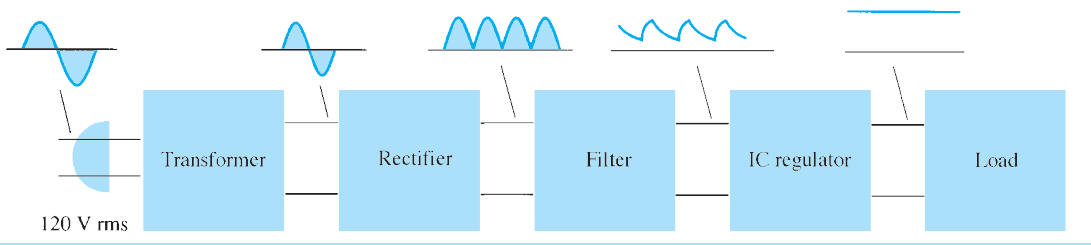


FIG. 15.1
Block diagram showing parts of a power supply.

15.2 GENERAL FILTER CONSIDERATIONS

A rectifier circuit is necessary to convert a signal having zero average value into one that has a nonzero average. The output resulting from a rectifier is a pulsating dc voltage and not yet suitable as a battery replacement. Such a voltage could be used in, say, a battery charger, where the average dc voltage is large enough to provide a charging current for the battery. For dc supply voltages, such as those used in a radio, stereo system, computer, and so on, the pulsating dc voltage from a rectifier is not good enough. A filter circuit is necessary to provide a steadier dc voltage.

Filter Voltage Regulation and Ripple Voltage

Before going into the details of a filter circuit, it would be appropriate to consider the usual methods of rating filter circuits so that we can compare a circuit's effectiveness as a filter. Figure 15.2 shows a typical filter output voltage, which will be used to define some of the signal factors. The filtered output of Fig. 15.2 has a dc value and some ac variation (ripple). Although a battery has essentially a constant or dc output voltage, the dc voltage derived from an ac source signal by rectifying and filtering will have some ac variation (ripple). The smaller the ac variation with respect to the dc level, the better is the filter circuit's operation.

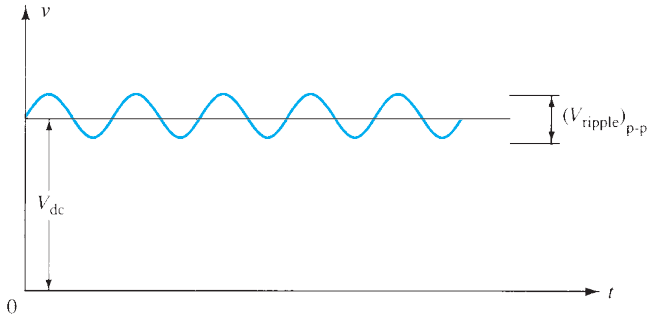


FIG. 15.2
Filter voltage waveform showing dc and ripple voltages.

Consider measuring the output voltage of a filter circuit using a dc voltmeter and an ac (rms) voltmeter. The dc voltmeter will read only the average or dc level of the output voltage. The ac (rms) meter will read only the rms value of the ac component of the output voltage (assuming the ac signal is coupled through a capacitor to block out the dc level).

Definition: Ripple is defined as

$$r = \frac{\text{ripple voltage (rms)}}{\text{dc voltage}} = \frac{V_r(\text{rms})}{V_{dc}} \times 100\% \tag{15.1}$$

EXAMPLE 15.1 Using a dc and ac voltmeter to measure the output signal from a filter circuit, we obtain readings of 25 V dc and 1.5 V rms. Calculate the ripple of the filter output voltage.

Solution:

$$r = \frac{V_r(\text{rms})}{V_{dc}} \times 100\% = \frac{1.5\text{ V}}{25\text{ V}} \times 100\% = \mathbf{6\%}$$

Voltage Regulation Another factor of importance in a power supply is the amount the dc output voltage changes over a range of circuit operation. The voltage provided at the

output under no-load condition (no current drawn from the supply) is reduced when load current is drawn from the supply (under load). The amount the dc voltage changes between the no-load and load conditions is described by a factor called voltage regulation.

Definition: Voltage regulation is given by

$$\text{Voltage regulation} = \frac{\text{no-load voltage} - \text{full-load voltage}}{\text{full-load voltage}}$$

$$\boxed{\%\text{V.R.} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\%} \tag{15.2}$$

EXAMPLE 15.2 A dc voltage supply provides 60 V when the output is unloaded. When connected to a load, the output drops to 56 V. Calculate the value of voltage regulation.

Solution:

$$\text{Eq. (15.2):} \quad \%\text{V.R.} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\% = \frac{60\text{ V} - 56\text{ V}}{56\text{ V}} \times 100\% = \mathbf{7.1\%}$$

If the value of full-load voltage is the same as the no-load voltage, the voltage regulation calculated is 0%, which is the best expected. This means that the supply is a perfect voltage source for which the output voltage is independent of the current drawn from the supply. The smaller the voltage regulation, the better is the operation of the voltage supply circuit.

Ripple Factor of Rectified Signal Although the rectified voltage is not a filtered voltage, it nevertheless contains a dc component and a ripple component. We will see that the full-wave rectified signal has a larger dc component and less ripple than the half-wave rectified voltage.

Half-wave: For a half-wave rectified signal, the output dc voltage is

$$V_{dc} = 0.318V_m \tag{15.3}$$

The rms value of the ac component of the output signal can be calculated (see Appendix C) to be

$$V_r(\text{rms}) = 0.385V_m \tag{15.4}$$

The percentage ripple of a half-wave rectified signal can then be calculated as

$$r = \frac{V_r(\text{rms})}{V_{dc}} \times 100\% = \frac{0.385V_m}{0.318V_m} \times 100\% = 121\% \tag{15.5}$$

Full-wave: For a full-wave rectified voltage the dc value is

$$V_{dc} = 0.636V_m \tag{15.6}$$

The rms value of the ac component of the output signal can be calculated (see Appendix C) to be

$$V_r(\text{rms}) = 0.308V_m \tag{15.7}$$

The percentage ripple of a full-wave rectified signal can then be calculated as

$$r = \frac{V_r(\text{rms})}{V_{dc}} \times 100\% = \frac{0.308V_m}{0.636V_m} \times 100\% = 48\% \tag{15.8}$$

In summary, a full-wave rectified signal has less ripple than a half-wave rectified signal and is thus better to apply to a filter.

15.3 CAPACITOR FILTER

A very popular filter circuit is the capacitor-filter circuit shown in Fig. 15.3. A capacitor is connected at the rectifier output, and a dc voltage is obtained across the capacitor. Figure 15.4a shows the output voltage of a full-wave rectifier before the signal is filtered, whereas Fig. 15.4b shows the resulting waveform after the filter capacitor is connected at the rectifier output. Notice that the filtered waveform is essentially a dc voltage with some ripple (or ac variation).

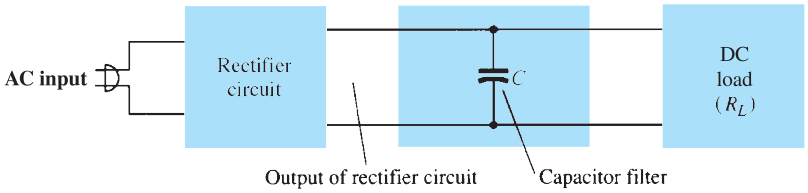


FIG. 15.3
Basic capacitor filter.

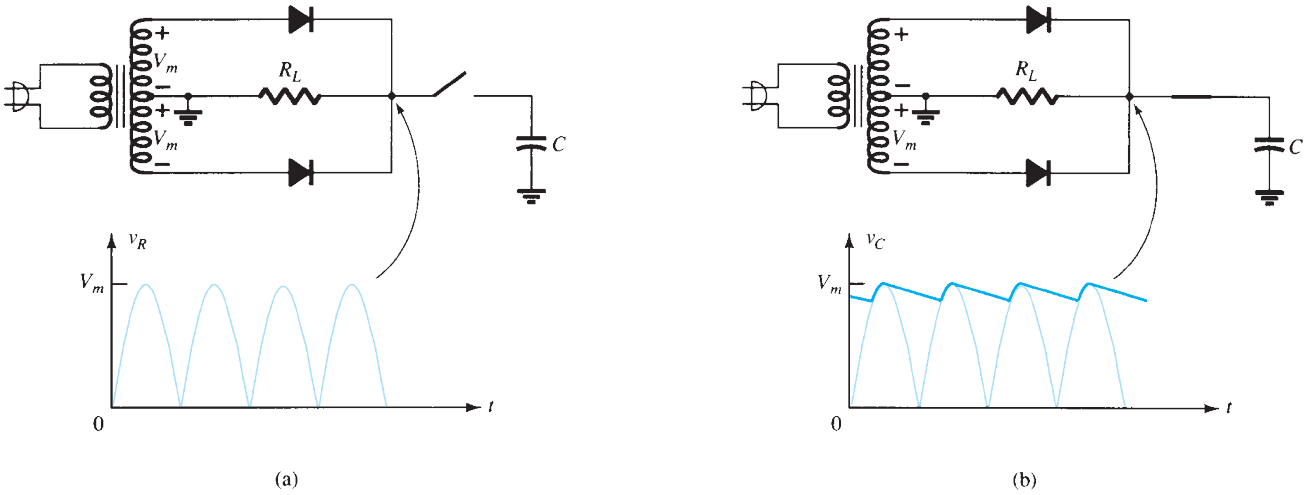


FIG. 15.4
Capacitor filter operation: (a) full-wave rectifier voltage; (b) filtered output voltage.

Figure 15.5a shows a full-wave bridge rectifier and the output waveform obtained from the circuit when connected to a load (R_L). If no load were connected across the capacitor, the output waveform would ideally be a constant dc level equal in value to the peak voltage (V_m) from the rectifier circuit. However, the purpose of obtaining a dc voltage is to provide

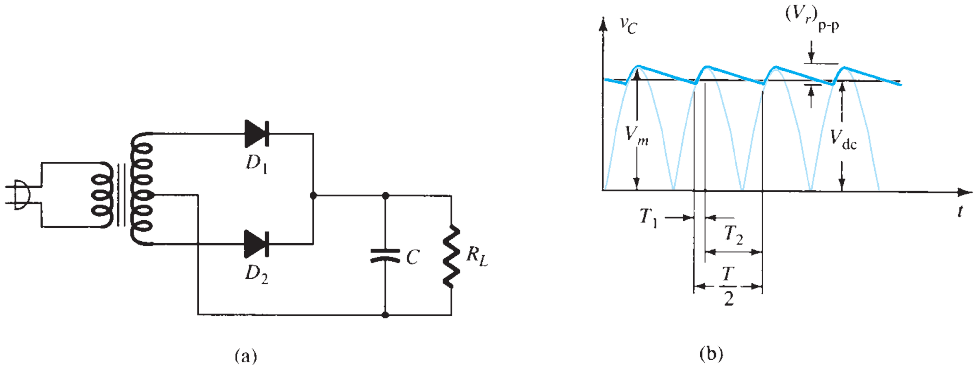


FIG. 15.5
Capacitor filter: (a) capacitor filter circuit; (b) output voltage waveform.

this voltage for use by various electronic circuits, which then constitute a load on the voltage supply. Since there will always be a load on the filter output, we must consider this practical case in our discussion.

Output Waveform

Figure 15.5b shows the waveform across a capacitor filter. Time T_1 is the time during which diodes of the full-wave rectifier conduct, charging the capacitor up to the peak rectifier voltage V_m. Time T_2 is the time interval during which the rectifier voltage drops below the peak voltage, and the capacitor discharges through the load. Since the charge–discharge cycle occurs for each half-cycle for a full-wave rectifier, the period of the rectified waveform is $T/2$. The filtered voltage, as shown in Fig. 15.6, shows the output waveform to have a dc level V_{dc} and a ripple voltage V_r (rms) as the capacitor charges and discharges. Some details of these waveforms and the circuit elements are considered next.

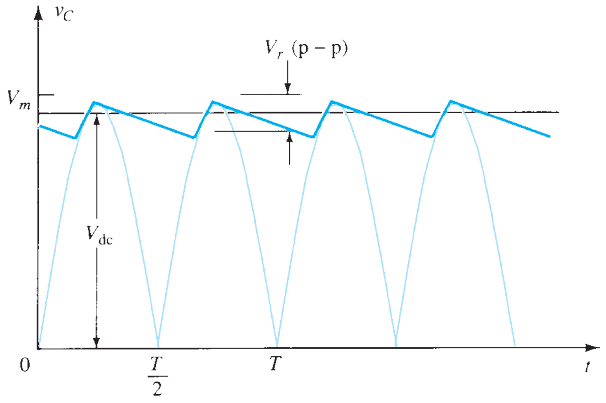


FIG. 15.6
Approximate output voltage of capacitor filter circuit.

Ripple Voltage V_r (RMS) Appendix C provides the details for determining the value of the ripple voltage in terms of the other circuit parameters. The ripple voltage can be calculated from

$$V_r(\text{rms}) = \frac{I_{\text{dc}}}{4\sqrt{3}fC} = \frac{2.4I_{\text{dc}}}{C} = \frac{2.4V_{\text{dc}}}{R_LC} \tag{15.9}$$

where I_{dc} is in milliamperes, C is in microfarads, and R_L is in kilohms.

EXAMPLE 15.3 Calculate the ripple voltage of a full-wave rectifier with a 100-μF filter capacitor connected to a load drawing 50 mA.

Solution:

$$\text{Eq. (15.9):} \quad V_r(\text{rms}) = \frac{2.4(50)}{100} = \mathbf{1.2\ V}$$

DC Voltage V_{dc} From Appendix C, we can express the dc value of the waveform across the filter capacitor as

$$V_{\text{dc}} = V_m - \frac{I_{\text{dc}}}{4fC} = V_m - \frac{4.17I_{\text{dc}}}{C} \tag{15.10}$$

where V_m is the peak rectifier voltage, I_{dc} is the load current in milliamperes, and C is the filter capacitor in microfarads.

EXAMPLE 15.4 If the peak rectified voltage for the filter circuit of Example 15.3 is 30 V, calculate the filter dc voltage.

Solution:

$$\text{Eq. (15.10):} \quad V_{dc} = V_m - \frac{4.17I_{dc}}{C} = 30 - \frac{4.17(50)}{100} = \mathbf{27.9\ V}$$

Filter Capacitor Ripple

Using the definition of ripple [Eq. (15.1)], Eq. (15.9), and Eq. (15.10), with $V_{dc} \approx V_m$, we can obtain the expression for the output waveform ripple of a full-wave rectifier and filter-capacitor circuit:

$$r = \frac{V_r(\text{rms})}{V_{dc}} \times 100\% = \frac{2.4I_{dc}}{CV_{dc}} \times 100\% = \frac{2.4}{R_L C} \times 100\% \qquad \textbf{(15.11)}$$

where I_{dc} is in milliamperes, C is in microfarads, V_{dc} is in volts, and R_L is in kilohms.

EXAMPLE 15.5 Calculate the ripple of a capacitor filter for a peak rectified voltage of 30 V, capacitor $C = 50\ \mu\text{F}$, and a load current of 50 mA.

Solution:

$$\text{Eq. (15.11):} \quad r = \frac{2.4I_{dc}}{CV_{dc}} \times 100\% = \frac{2.4(50)}{100(27.9)} \times 100\% = \mathbf{4.3\%}$$

We could also calculate the ripple using the basic definition:

$$r = \frac{V_r(\text{rms})}{V_{dc}} \times 100\% = \frac{1.2\ \text{V}}{27.9\ \text{V}} \times 100\% = \mathbf{4.3\%}$$

Diode Conduction Period and Peak Diode Current

From the previous discussion, it should be clear that larger values of capacitance provide less ripple and higher average voltage, thereby providing better filter action. From this one might conclude that to improve the performance of a capacitor filter it is only necessary to increase the size of the filter capacitor. The capacitor, however, also affects the peak current drawn through the rectifying diodes, and, as will be shown next, the larger the value of the capacitor, the larger is the peak current drawn through the rectifying diodes.

Recall that the diodes conduct during period T_1 (see Fig. 15.5), during which time the diode must provide the necessary average current to charge the capacitor. The shorter this time interval, the larger is the amount of the charging current. Figure 15.7 shows this relation for a half-wave rectified signal (it would be the same basic operation for the full-wave case). Notice that for smaller values of capacitor, with T_1 larger, the peak diode current is less than for larger values of filter capacitor.

Since the average current drawn from the supply must equal the average diode current during the charging period, the following relation can be used (assuming constant diode current during charge time):

$$I_{dc} = \frac{T_1}{T} I_{peak}$$

from which we obtain

$$I_{peak} = \frac{T}{T_1} I_{dc} \qquad \textbf{(15.12)}$$

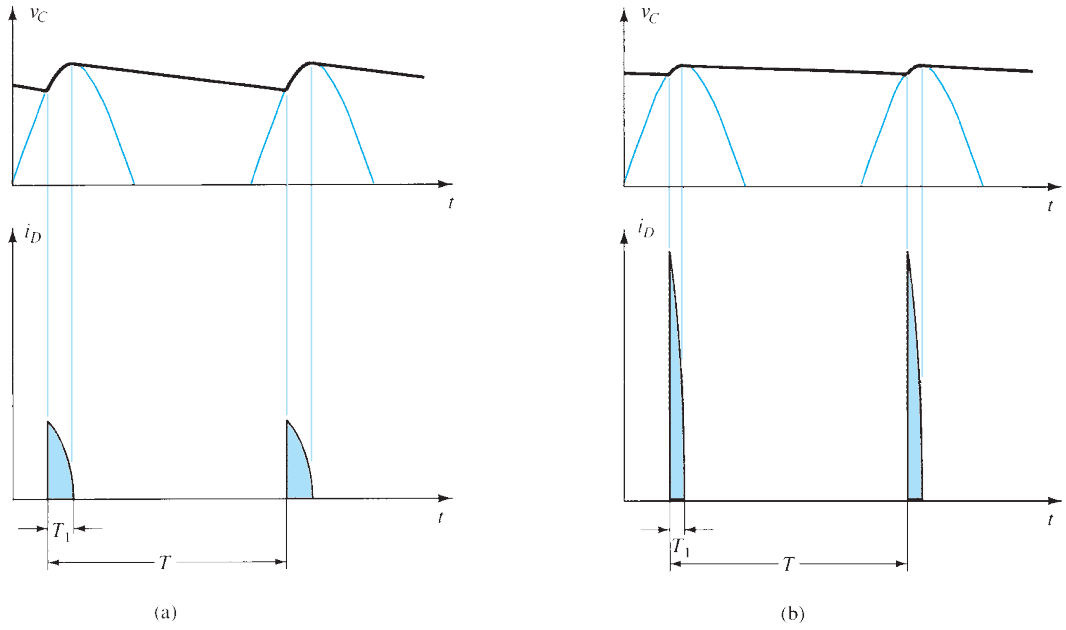


FIG. 15.7
Output voltage and diode current waveforms: (a) small C; (b) large C.

where T_1 = diode conduction time
$T = 1/f$ ($f = 2 \times 60$ for the full-wave case)
I_{dc} = average current drawn from the filter
I_{peak} = peak current through the conducting diodes

15.4 *RC* FILTER

It is possible to further reduce the amount of ripple across a filter capacitor by using an additional *RC* filter section as shown in Fig. 15.8. The purpose of the added *RC* section is to pass most of the dc component while attenuating (reducing) as much of the ac component as possible. Figure 15.9 shows a full-wave rectifier with capacitor filter followed by an *RC* filter section. The operation of the filter circuit can be analyzed using superposition for the dc and ac components of the signal.

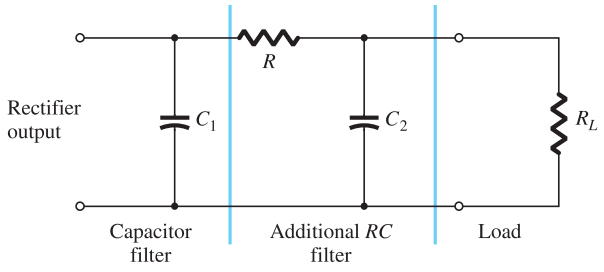


FIG. 15.8
RC filter stage.

DC Operation of *RC* Filter Section

Figure 15.10a shows the dc equivalent circuit to use in analyzing the *RC* filter circuit of Fig. 15.9. Since both capacitors are open-circuit for dc operation, the resulting output dc voltage is

$$V'_{dc} = \frac{R_L}{R + R_L} V_{dc} \tag{15.13}$$

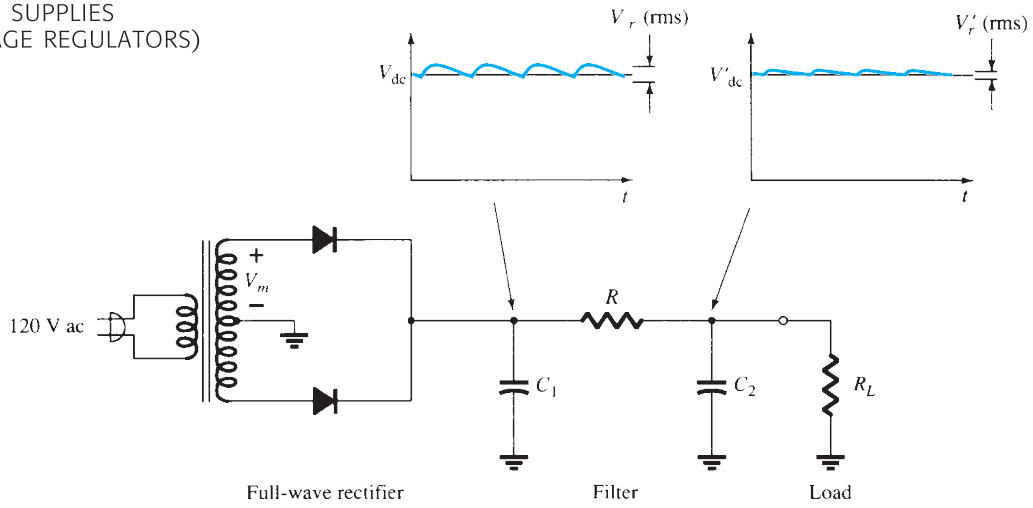


FIG. 15.9
Full-wave rectifier and RC filter circuit.

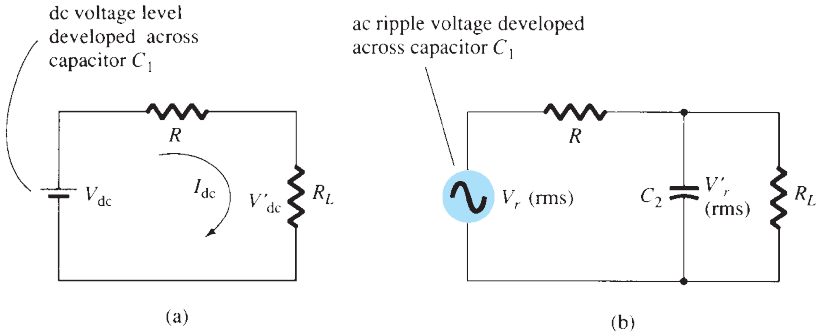


FIG. 15.10
(a) DC and (b) ac equivalent circuits of RC filter.

EXAMPLE 15.6 Calculate the dc voltage across a 1-kΩ load for an *RC* filter section ($R = 120\ \Omega$, $C = 10\ \mu$F). The dc voltage across the initial filter capacitor is $V_{dc} = 60$ V.

Solution:

$$\text{Eq. (15.13):}\quad V'_{dc} = \frac{R_L}{R + R_L} V_{dc} = \frac{1000}{120 + 1000}(60\text{ V}) = \mathbf{53.6\ V}$$

AC Operation of *RC* Filter Section

Figure 15.10b shows the ac equivalent circuit of the *RC* filter section. Due to the voltage-divider action of the capacitor ac impedance and the load resistor, the ac component of voltage resulting across the load is

$$V'_r(\text{rms}) \approx \frac{X_C}{R} V_r(\text{rms}) \qquad \textbf{(15.14)}$$

For a full-wave rectifier with ac ripple at 120 Hz, the impedance of a capacitor can be calculated using

$$X_C = \frac{1.3}{C} \qquad \textbf{(15.15)}$$

where C is in microfarads and X_C is in kilohms.

EXAMPLE 15.7 Calculate the dc and ac components of the output signal across load R_L in the circuit of Fig. 15.11. Calculate the ripple of the output waveform.

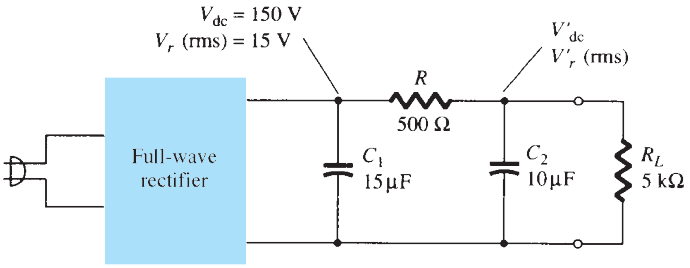


FIG. 15.11
RC filter circuit for Example 15.7.

Solution:

DC Calculation We obtain

$$\text{Eq. (15.13):} \quad V'_{dc} = \frac{R_L}{R + R_L} V_{dc} = \frac{5\,\text{k}\Omega}{500 + 5\,\text{k}\Omega}(150\text{ V}) = \mathbf{136.4\text{ V}}$$

AC Calculation The *RC*-section capacitive impedance is

$$\text{Eq. (15.15):} \quad X_C = \frac{1.3}{C} = \frac{1.3}{10} = 0.13\,\text{k}\Omega = 130\,\Omega$$

The ac component of the output voltage, calculated using Eq. (15.14), is

$$V'_r(\text{rms}) = \frac{X_C}{R} V_r(\text{rms}) = \frac{130}{500}(15\text{ V}) = \mathbf{3.9\text{ V}}$$

The ripple of the output waveform is then

$$r = \frac{V'_r(\text{rms})}{V'_{dc}} \times 100\% = \frac{3.9\text{ V}}{136.4\text{ V}} \times 100\% = \mathbf{2.86\%}$$

15.5 DISCRETE TRANSISTOR VOLTAGE REGULATION

Two types of transistor voltage regulators are the series voltage regulator and the shunt voltage regulator. Each type of circuit can provide an output dc voltage that is regulated or maintained at a set value even if the input voltage varies or if the load connected to the output changes.

Series Voltage Regulation

The basic connection of a series regulator circuit is shown in the block diagram of Fig. 15.12. The series element controls the amount of the input voltage that gets to the output.

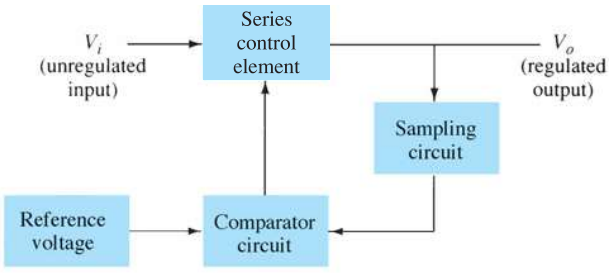


FIG. 15.12
Series regulator block diagram.

The output voltage is sampled by a circuit that provides a feedback voltage to be compared to a reference voltage.

1. If the output voltage increases, the comparator circuit provides a control signal to cause the series control element to decrease the amount of the output voltage—thereby maintaining the output voltage.
2. If the output voltage decreases, the comparator circuit provides a control signal to cause the series control element to increase the amount of the output voltage.

Series Regulator Circuit A simple series regulator circuit is shown in Fig. 15.13. Transistor Q_1 is the series control element, and Zener diode D_Z provides the reference voltage. The regulating operation can be described as follows:

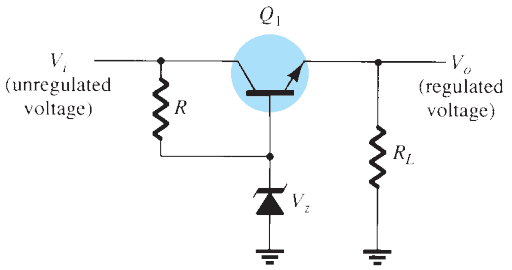


FIG. 15.13
Series regulator circuit.

1. If the output voltage decreases, the increased base-emitter voltage causes transistor Q_1 to conduct more, thereby raising the output voltage—maintaining the output constant.
2. If the output voltage increases, the decreased base-emitter voltage causes transistor Q_1 to conduct less, thereby reducing the output voltage—maintaining the output constant.

EXAMPLE 15.8 Calculate the output voltage and the Zener current in the regulator circuit of Fig. 15.14 for $R_L = 1\ \text{k}\Omega$.

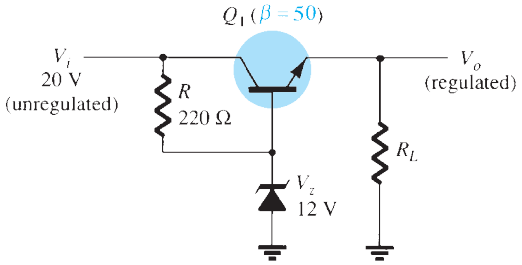


FIG. 15.14
Circuit for Example 15.8.

Solution:

$$V_o = V_Z - V_{BE} = 12\ \text{V} - 0.7\ \text{V} = \mathbf{11.3\ V}$$

$$V_{CE} = V_i - V_o = 20\ \text{V} - 11.3\ \text{V} = 8.7\ \text{V}$$

$$I_R = \frac{20\ \text{V} - 12\ \text{V}}{220\ \Omega} = \frac{8\ \text{V}}{220\ \Omega} = 36.4\ \text{mA}$$

For $R_L = 1\ \text{k}\Omega$,

$$I_L = \frac{V_o}{R_L} = \frac{11.3\ \text{V}}{1\ \text{k}\Omega} = 11.3\ \text{mA}$$

$$I_B = \frac{I_C}{\beta} = \frac{11.3\text{ mA}}{50} = 226\ \mu\text{A}$$

$$I_Z = I_R - I_B = 36.4\text{ mA} - 226\ \mu\text{A} \approx \mathbf{36\ mA}$$

Improved Series Regulator An improved series regulator circuit is shown in Fig. 15.15. Resistors R_1 and R_2 act as a sampling circuit, with Zener diode D_Z providing a reference voltage, and transistor Q_2 then controls the base current to transistor Q_1 to vary the current passed by transistor Q_1 to maintain the output voltage constant.

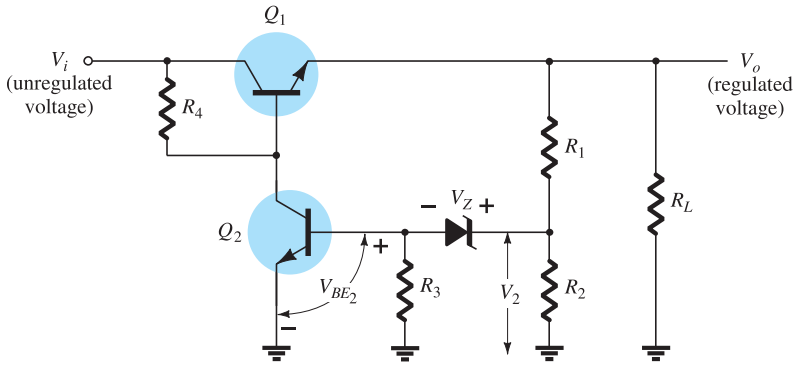


FIG. 15.15
Improved series regulator circuit.

If the output voltage tries to increase, the increased voltage, V_2, sampled by R_1 and R_2, causes the base-emitter voltage of transistor Q_2 to go up (since V_Z remains fixed). If Q_2 conducts more current, less goes to the base of transistor Q_1, which then passes less current to the load, reducing the output voltage—thereby maintaining the output voltage constant. The opposite takes place if the output voltage tries to decrease, causing less current to be supplied to the load, to keep the voltage from decreasing.

The voltage V_2 provided by sensing resistors R_1 and R_2 must equal the sum of the base-emitter voltage of Q_2 and the Zener diode, that is,

$$V_{BE_2} + V_Z = V_2 = \frac{R_2}{R_1 + R_2} V_o \qquad \textbf{(15.16)}$$

Solving Eq. (15.16) for the regulated output voltage V_o gives

$$\boxed{V_o = \frac{R_1 + R_2}{R_2}(V_Z + V_{BE_2})} \qquad \textbf{(15.17)}$$

EXAMPLE 15.9 What regulated output voltage is provided by the circuit of Fig. 15.15 for the circuit elements $R_1 = 20\text{ k}\Omega$, $R_2 = 30\text{ k}\Omega$, and $V_Z = 8.3$ V?

Solution: From Eq. (15.17), the regulated output voltage is

$$V_o = \frac{20\text{ k}\Omega + 30\text{ k}\Omega}{30\text{ k}\Omega}(8.3\text{ V} + 0.7\text{ V}) = \mathbf{15\ V}$$

Op-Amp Series Regulator Another type of series regulator is shown in Fig. 15.16. The op-amp compares the Zener diode reference voltage with the feedback voltage from sensing resistors R_1 and R_2. If the output voltage varies, the conduction of transistor Q_1 is controlled to maintain the output voltage constant. The output voltage will be maintained at a value of

$$\boxed{V_o = \left(1 + \frac{R_1}{R_2}\right)V_Z} \qquad \textbf{(15.18)}$$